

Sleep, Athletic Performance, and Recovery

17-22 minutes : 1/22/2021

DOI: <https://doi.org/10.1016/j.physbeh.2015.08.035>.

We Value Your Privacy

We use cookies to improve your experience on our website, analyze site usage, and provide you with tailored content. By clicking 'Accept All Cookies', you agree to our use of cookies. You can manage your cookie preferences by clicking 'Manage Cookies'.

[Skip to content](#)

ADVERTISEMENT

Ads on our site helps support our mission to improve sleep health. Sleep Foundation doesn't endorse third-party products or services. [Policy](#)

Athletes know that [physical activity](#) is an important component of a healthy lifestyle. Regular exercise increases longevity and reduces the risks of cardiovascular disease, diabetes, and some cancers. Exercise can also reduce the risk for anxiety and depression, and it [can help you sleep better](#).

In order to perform their best, athletes must prepare in every aspect of their lives. They train regularly, eat healthy meals and snacks, and make time for rest, recovery, and sleep. When one area is lacking, overall performance can suffer. Sleep is certainly no exception!

For both athletes and non-athletes, [sleep is essential](#) for overall health and wellbeing. Everyone needs sleep in order to feel restored and function their best the next day. Other physical benefits include:

- **Allowing your** heart to rest and cells and tissue to repair. This can help your body recover after physical exertion. Also, as you progress through the stages of sleep, the changes in your heart rate and breathing throughout the night promote cardiovascular health.
- **Preventing illness** or helping you recover from illness. During sleep, your body produces *cytokines*, which are hormones that help the immune system fight off infections.

All of these restorative effects are important for athletes' recovery and performance.



How Sleep Helps an Athlete's Mental State

Sleep helps everyone to retain and consolidate memories. When athletes practice or learn new skills, sleep helps form memories, and contributes to improved performance in the future. Without sleep, the pathways in the brain that allow you to learn and make memories can't be formed or maintained.

Sleep is also essential for cognitive processing. Loss of sleep is associated with a decline in cognitive function. This can have adverse effects on athletes whose sports require a high level of cognitive function, such as decision making and adapting to new situations.

Also, just as exercise can help improve or maintain mental health, sleep is important for maintaining athlete's mental health. Quality sleep is associated with improving overall mood. Healthy sleep prevents irritability and decreases the risk of developments such as depression.

Both increased quantity and quality of sleep helps athletes improve performance in many areas related to the demands of the sport.

- A Stanford study of men's basketball players who extended their sleep to 10 hours a night found several positive outcomes. The players ran faster in both half-court and full-court sprints. Their shooting improved by at least 9% for both free throws and three-point shots. The athletes also reported improved physical and mental well-being.
- Male and female swimmers who extended their sleep to 10 hours also saw many performance improvements. Reaction times off diving blocks were faster, turn times were improved, and kick strokes increased. Times swimming a 15-meter sprint also improved. Additionally, these athletes experienced improved mood and decreased daytime sleepiness and fatigue.

- Varsity tennis players, male and female, who increased their sleep to at least nine hours a week also performed better. The accuracy of the players' serves increased significantly from about 36% to nearly 42%. The players experienced less sleepiness as well.
- Other studies of female netball players and male soccer players have demonstrated that sleep hygiene education helps athletes increase their overall sleep time. This adequate sleep before a competition is likely to encourage top performance.

A Lack of Sleep Affects an Athlete's Performance

Poor quality and quantity of sleep lead to several negative effects in any person. Mentally, sleep deprivation reduces the ability to react quickly and think clearly. People who are sleep deprived are more likely to make poor decisions and take risks. A lack of sleep also increases irritability and risk for anxiety and depression. Physically, a lack of sleep increases the risk for many medical concerns, including type 2 diabetes, high blood pressure, kidney disease, and stroke.

While quality sleep has positive effects specifically on athletic performance, a lack of sleep is detrimental to performance. A great number of concerns can arise when athletes do not receive adequate sleep:

- **Inhibited ability.** In a study of male team-sport athletes who were sleep-deprived, average and total sprint times decreased.
- **Decreased accuracy.** After sleep deprivation, male and female tennis players had decreased serve accuracy of up to 53% when compared to performance after normal sleep.
- **Quicker exhaustion.** In a study of male runners and volleyball players, both groups of athletes exhausted faster after sleep deprivation.
- **Decreased reaction time.** A lack of sleep adversely affected reaction time in a studied group of male collegiate athletes.
- **Difficulty learning and decision making.** Executive functions are impacted by a lack of sleep. Choices such as passing the ball or taking it to the net yourself can be more difficult or made too late.
- **Risk for injury.** Research of middle and high school athletes revealed that a chronic lack of sleep is associated with increased rates of injury.
- **Risk for illness or immunosuppression.** Poor sleep habits are associated with lower resistance to illness, such as the common cold.

Sleep Better With Our Top Sleep Tips

Sign up to receive emails about how to improve your sleep, as well as our comprehensive Better Sleep Guidebook.



Evidence shows that more sleep, or extended sleep, can benefit athletes, their recovery, and their performance. Recommendations for athletes range between seven and nine hours nightly. Elite athletes are encouraged to get at least nine hours of sleep nightly and to treat sleep with as much importance as athletic training and diet. In contrast, people who exercise moderately likely do not need as much sleep as elite performers. [Standard sleep guidelines](#) are appropriate.

While it is not recommended for some sleepers, such as those with insomnia, napping after a night of inadequate sleep can benefit athletes. Athletes who anticipate a night of inadequate sleep can also benefit from extending their sleep in the nights beforehand. Additional sleep is encouraged before events such as traveling to competitions, before a heavy competition, and during times of illness or injury.

For some types of athletes, waking early has more of a negative impact than staying up late. A study of judo athletes showed that sleep deprivation at the end of the sleeping time (i.e., early morning) decreased power and muscle strength the following day. If early wake times are affecting your performance, consider consulting your coach to determine a training and competition schedule that best meets your needs.

Stages of Sleep for Athletes

Different functions happen throughout each of the [stages of sleep](#), and all are necessary in order to have healthy sleep. But are there any parts of the sleep cycle that are particularly beneficial to athletes?

The results of a study of Norwegian chess players suggest so. Of the players studied, those who improved their chess ranking had different sleep patterns from the players whose chess rankings dropped. The sleep patterns of the improved players had less rapid eye movement (REM) sleep, higher amounts of deep sleep, and lower respiration rates.

Sleep hygiene is important for all people to sleep well. Common components include:

- **Creating an appropriate sleep environment.** Your sleeping space should be dark and cool with little to no noise. Your sleep environment should be used only for sex and sleep.
- **Avoid alcohol and caffeine before bedtime.** These beverages can interrupt sleep or lead to more disturbed sleep.

- **Stay away from electronics in the hours before bedtime.** This includes TVs, cell phones, and computers. The blue light that these devices emit can affect your circadian rhythm.
- **Have a wind-down routine.** Activities such as reading, taking a bath, or meditating can help you relax and get ready for sleep.
- **Get out of bed if you can't fall asleep after 20 minutes of trying.** Do a quiet activity in another space until you feel sleepy.

In addition to these sleep hygiene tips, other habits especially important for athletes are to:

- **Avoid overtraining.** Keep a consistent training schedule so as not to overexert yourself.
- **Avoid training and competitions too early or too late.** These can affect sleep quantity and quality, especially if your athletic schedule is inconsistent.
- **Keep naps brief, if you take them at all.** Naps should be no more than an hour and not taken after 3 p.m..
- **Reduce stressors.** Not only do mental stressors affect sleep quality, but they also impact performance overall.

Jet Lag in Athletes

Another aspect of sleep quality athletes need to consider is the effects of jet lag. When traveling to different time zones for competitions, athletes can get out of their natural circadian phase. This means athletes may experience fatigue or the inability to perform their best. For example, West Coast American football teams play significantly better during evening home games than the visiting East Coast teams.

To combat the negative effects of jet lag, athletes should consider additional sleep hygiene tips for travel:

- **Prepare for travel.** Adjust your sleep schedule to mimic the time of the destination you're traveling to in order to quickly adjust to the time upon arrival. Set your watch for the destination time zone when you board the plane.
- **Get enough sleep before traveling.** To avoid sleep debt upon arrival, be sure to sleep prior to and during travel, if necessary.
- **Make a comfortable environment.** Pillows can be used for cushion and comfort. Earplugs and eye masks can help create a quiet, dark environment for sleep during a flight. Avoid distractions such as electronics.
- **Stay hydrated.** On the airplane, be sure to take plenty of fluids. Avoid caffeine and alcohol.
- **Eat meals on the destination time.** Arranging meal-times according to your destination time-zone can also shorten the time it takes to adjust to a new time zone.